

PART III

PLANNING YOUR TRAINING



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TRAINING SESSION DESIGN



Now you know what to train and how to train it, you need to bring together individual exercises, each type of training, into the basic unit of any training plan: a training session. You could work on one area or skill per session, or focus on multiple skills in the same session. However, if you train more than one skill per session, bear in mind that the order of these skills will change the end product. What's more, while there's a positive transfer between certain skills, some have a negative transfer and others can lead to training interference.

SINGLE-FOCUS SESSIONS. EXAMPLE SESSION

The first thing to decide is what to work on, then you need to decide how to work on it. You could do isolated exercises (for example, deadhangs to build maximum finger strength), climbing exercises (crimpy problems), or a combination of both. This choice will depend on your level of ability, training goals, strengths and weaknesses, and where you are in your training cycle.

Let's use the following model for the example session in this chapter:

1. Warm-up:

- *General*: joint mobility, increasing ROM, muscle activation, etc.
- *Specific/wall-based*: traversing, introducing the target technique, core, etc.
- *For more advanced levels*: PAP exercises.

2. Main part of the session:

- *Technique*: it's essential to train technique **without fatigue and at the actual intensity** at which you'd use the technique on a boulder problem, traverse, pitch, etc.
 - a) Climbing exercises:
 - Boulder problems or routes featuring the skill that you want to work on (crimps, dynos, campusing, slopers, endurance, continuity), using the most suitable method for what it is that you want to train.
 - *Complementary isolated exercises*: depending on climbing ability, training needs and how much time you have left, you could also do some isolated exercises (such as deadhangs or pull-ups), **as long as you're able to maintain the right level of intensity** for your chosen exercise. For example, if you're tired and can't move at maximum speed, campusing would be a bad idea as you'd not produce the necessary power/RFD and you'd also be increasing your risk of injury.
 - *Antagonist training*: end your session with some antagonist training for injury prevention, activating and strengthening the antagonist muscles to balance out their agonist counterparts.
 - b) Isolated exercises:
 - *Select the relevant exercises and methods*: deadhangs, pull-ups, lock-offs, campusing, etc. Remember that this type of session is for more advanced climbers looking to isolate specific aspects of physical conditioning for climbing.
 - *Antagonist training*: end your session with some antagonist training for injury prevention, activating and strengthening the antagonist muscles to balance out their agonist counterparts.
 - c) Isolated exercises + climbing exercises (intra-session transfer):
 - *Isolated exercises*: do your isolated exercises first to make sure you can maintain the right level of intensity.
 - *Complementary physical conditioning*: as you'll need a good amount of rest to ensure good technique and the right level of intensity in your climbing exercises, you could use this rest time for some physical conditioning. Focus on muscles which are often neglected or which might need a little extra attention (for example, hamstring strength for more powerful heel hooks).
 - *Climbing exercises*: this is when to focus on the inter-session transfer of skills. For example, in a finger strength training session, you could do a range of crimpy boulder problems after your max-weight deadhangs.

The following example is a maximum strength training session with both isolated exercises and climbing exercises for an intermediate/advanced climber:

Warm-up:

1. *Joint mobility*: 10 seconds per joint, from ankles to wrists/fingers.
2. *Cardio*: two minutes of skipping at an intensity of 3–4/10.
3. *Increasing ROM*: foam roll calves, hamstrings, glutes, back and between the shoulder blades. Hip abduction with local vibration on gluteus medius, pushing against a wall for resistance: five reps with five seconds of isometric contractions. Knee raises with local vibration on hip flexors (psoas and rectus femoris).
4. *Muscle activation*: two sets of: pull-ups (6 reps) + Bulgarian lunge (10 reps) + forearm plank (3 x 8 seconds x 2 seconds rest) + military press with resistance band (10 reps) + glute bridge (15 reps) + ab roll-out (8 reps) + biceps and triceps with resistance band (12 reps each). Rest for one minute between sets.
5. *Traversing*: four traverses with 40–30–20–15 moves. Incremental intensity: do the first three traverses on a vertical wall with the 20-move traverse on one-pad holds, then do the fourth traverse on a slightly overhanging wall also on one-pad holds. Rest for 30 to 45 seconds between each traverse.
6. *Wall-based core work*: feet forwards. Three sets of four to five moves, resting for one minute between sets.

Main part of the session:

1. *Deadhangs*:
Preparation: 3 hangs x 20mm – 16mm – 14mm x 8 seconds: 1 minute of rest Five max-weight hangs x 18mm x 8 seconds: 3–5 minutes of rest
2. *Complementary physical conditioning*:
Two sets x modified razor curl with hands on chest x 8 reps x 90 seconds of rest Four sets x Nordic curl x 5 reps x 2 minutes of rest

3. *Bouldering:*

Pick a five-move crimping overhanging problem, EL 5(6). Rest for three minutes between each attempt. Have as many attempts as possible. End the exercise when you start falling, due to tiredness, on move three or lower. Focus on problems with lock-offs and heel hooks.

Cool down:

1. *Traversing:* two to three traverses on a vertical wall, 15 to 20 moves, intensity 3–4/10, using minimum two-pad holds. Rest for 30 to 60 seconds.

CRITERIA FOR PLANNING A MULTI-FOCUS SESSION. TRANSFER AND INTERFERENCE

If you want to work on multiple areas in one session, it's important to consider the following points:

- Effective maximum strength training requires rest. If you don't rest for long enough between sets and exercises, neither the nervous system (neurotransmitter replenishment) nor the metabolism (phosphagen replenishment) will be in the best condition for the next set or exercise. As such, you won't be able to perform at your best and will end up with fatigue instead of the intended training stimulus.
- Effective power and RFD training requires high levels of strength. You should never train power or RFD to exhaustion. Remember that if you're tired and can't produce enough force, the end product (power) will be lower and you won't achieve the intended training stimulus.
- Endurance training interferes with strength gains. Strength training activates the mTOR pathway (a series of metabolic chemical reactions involving the secretion of certain hormones and anabolic processes responsible for building up organic structures, among other physiological functions), while endurance training activates the AMPK pathway (a series of metabolic chemical reactions involving the secretion of certain hormones and catabolic processes responsible for breaking down organic structures in order to produce energy, among other physiological functions). They are opposing and incompatible pathways (adapted from González-Badillo and Gorostiaga-Ayestarán, 2002). However, emerging evidence suggests that this interference can be reduced to a minimum if certain recovery times (specific to the training method and intensity) are observed.

With this in mind, some potential combinations include:

- *Maximum strength sessions:*
 - a) Maximum finger strength + maximum pulling strength
 - b) Maximum finger strength/pulling strength + power/RFD – only if strength and speed aren't limited by fatigue and only in this order (not power/RFD + max strength)
- *Power/RFD sessions:*
 - a) Single focus
- *Endurance sessions:*
 - a) Single focus
 - b) Endurance + antagonist strength/non-targeted muscle groups: These sessions can be used to work on areas of weakness and to build antagonist strength for injury prevention.